

ILHEUS, Brazil

WMO: 833490

Lat: 14.816S

Lon: 39.033W

Elev: 5

StdP: 101.27

Time Zone: -3.00 (W03)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	19.0	16.0	11.4	21.3	16.9	12.1	21.8	9.1	24.2	8.3	24.4	3.2	270	0.335

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.0	31.1	26.1	30.6	25.8	30.1	25.5	26.7	29.6	26.5	29.4	26.2	29.0	4.2	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	28.7	25.5	20.8	28.4	25.2	20.3	28.1	84.1	29.6	82.5	29.6	81.2	29.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.6	6.6	6.0	DB	16.9	31.4	1.0	0.9	16.2	32.0	15.6	32.6	15.1	33.1	14.4	33.8
			WB	16.1	27.6	0.9	1.0	15.5	28.3	14.9	28.9	14.4	29.4	13.8	30.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	26.7	26.8	26.8	26.2	25.1	24.0	23.2	23.1	23.9	25.0	25.7	26.3	(6)
	DBStd	1.72	0.95	0.95	0.98	1.03	1.08	0.97	1.04	1.00	1.28	1.37	1.33	0.98	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	5555	516	469	521	487	467	420	409	408	416	466	471	507	(10)
	CDD18.3	2513	258	236	262	237	208	170	151	149	166	207	221	248	(11)
	CDH23.3	20517	2675	2456	2694	2089	1408	887	654	699	1021	1602	1921	2410	(12)
(13)	CDH26.7	4526	741	742	836	565	258	88	34	29	84	221	359	568	(13)
(14)	Wind	WSAvg	3.3	3.3	3.1	3.0	3.2	3.4	3.3	3.5	3.6	3.6	3.3	3.1	(14)
(15) Precipitation	PrecAvg	1615	138	148	173	152	120	144	147	107	95	116	157	120	(15)
	PrecMax	2529	366	342	409	352	379	488	339	227	211	269	322	352	(16)
	PrecMin	985	38	57	54	9	2	13	66	27	28	26	11	13	(17)
	PrecStd	381	75	76	92	84	87	94	62	51	44	57	79	73	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	31.2	31.8	31.2	30.1	29.1	28.2	28.1	29.1	29.9	30.2	31.1	(19)
		MCWB	26.0	26.4	26.3	26.4	25.6	24.3	23.6	23.3	24.3	25.2	25.4	25.6	(20)
	2%	DB	30.2	30.9	31.1	30.5	29.2	28.2	27.4	27.2	28.2	29.0	29.8	30.1	(21)
		MCWB	25.5	26.2	26.1	26.0	24.9	23.7	23.0	22.7	23.8	24.6	25.1	25.4	(22)
	5%	DB	30.0	30.2	30.2	30.0	28.9	27.8	27.0	27.0	27.8	28.2	29.1	29.5	(23)
		MCWB	25.3	25.6	25.7	25.6	24.6	23.5	22.7	22.6	23.5	24.1	24.7	25.1	(24)
	10%	DB	29.5	29.8	29.9	29.2	28.1	27.0	26.2	26.2	27.0	27.9	28.5	29.0	(25)
		MCWB	25.1	25.4	25.5	25.2	24.0	23.1	22.4	22.3	23.0	23.9	24.5	24.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	27.2	27.2	27.0	26.4	25.2	24.9	24.4	25.0	26.0	26.2	26.7	(27)
		MCDB	29.5	30.1	30.0	30.3	29.0	27.8	27.2	26.6	28.0	28.3	28.7	28.9	(28)
	2%	WB	26.4	26.6	26.7	26.5	25.7	24.6	23.9	23.7	24.5	25.4	25.7	26.2	(29)
		MCDB	29.2	29.7	29.7	29.6	28.4	27.2	26.4	26.3	27.4	27.9	28.3	28.6	(30)
	5%	WB	26.0	26.2	26.4	26.1	25.3	24.1	23.4	23.2	24.0	24.9	25.4	25.8	(31)
		MCDB	28.9	29.3	29.3	29.1	27.9	26.8	26.0	26.0	26.9	27.6	28.0	28.3	(32)
	10%	WB	25.6	25.8	26.0	25.7	24.7	23.6	22.9	22.7	23.6	24.5	25.0	25.5	(33)
		MCDB	28.6	28.9	29.0	28.5	27.2	26.2	25.5	25.5	26.4	27.2	27.7	28.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.0	6.0	5.9	6.0	6.1	6.2	6.4	6.0	5.3	5.2	5.3	(35)
		MCDBR	6.0	6.3	6.4	6.5	6.6	6.8	6.8	6.8	6.3	5.5	5.5	5.9	(36)
	5% WB	MCWBR	2.6	2.6	2.7	2.9	3.1	3.3	3.4	3.4	3.0	2.6	2.6	2.5	(37)
		MCWBR	5.6	5.9	6.0	6.0	6.2	6.2	6.3	6.3	5.8	5.0	5.1	5.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.394	0.389	0.390	0.387	0.381	0.369	0.361	0.371	0.423	0.427	0.421	0.412	(40)
		taud	2.503	2.524	2.536	2.545	2.552	2.572	2.579	2.524	2.367	2.381	2.422	2.445	(41)
	Ebn at Noon		950	947	925	894	866	860	878	897	881	900	918	932	(42)
		Edh at Noon	115	112	107	101	94	89	90	101	125	128	124	122	(43)
(44) All-Sky Solar Radiation	RadAvg		6.41	6.24	5.72	5.06	4.33	4.00	4.25	4.88	5.34	5.74	5.73	6.15	(44)
	RadStd		0.48	0.46	0.41	0.40	0.20	0.20	0.16	0.23	0.37	0.55	0.49	0.52	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (1 neighbor)	N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page